

ReLU Variants: Leaky ReLU, PReLU, ELU and SELU

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Vanishing gradient ([Note DL-018](#)); Activation functions ([Note DL-027](#)).

1. Overview

Key point: A ReLU node whose weighted sum stays negative outputs 0 with slope 0, so it never learns again: it is dead. Four variants give negative inputs a non-zero output and slope, which keeps every node alive.

ReLU (G-1668) is the default **activation function** (G-165) for **hidden layers** (G-890; see the [activation functions Note](#)). ReLU's biggest weakness is the **dying ReLU problem** (G-650). This Note covers:

- what a dead node is, why it happens and why it is permanent;
- three fixes, the third being a change of activation;
- two **linear variants** (G-1098; Leaky ReLU, Parametric ReLU), which change ReLU's negative side by a straight line;
- two **non-linear variants** (G-1336; ELU, SELU), which use an exponential curve there.

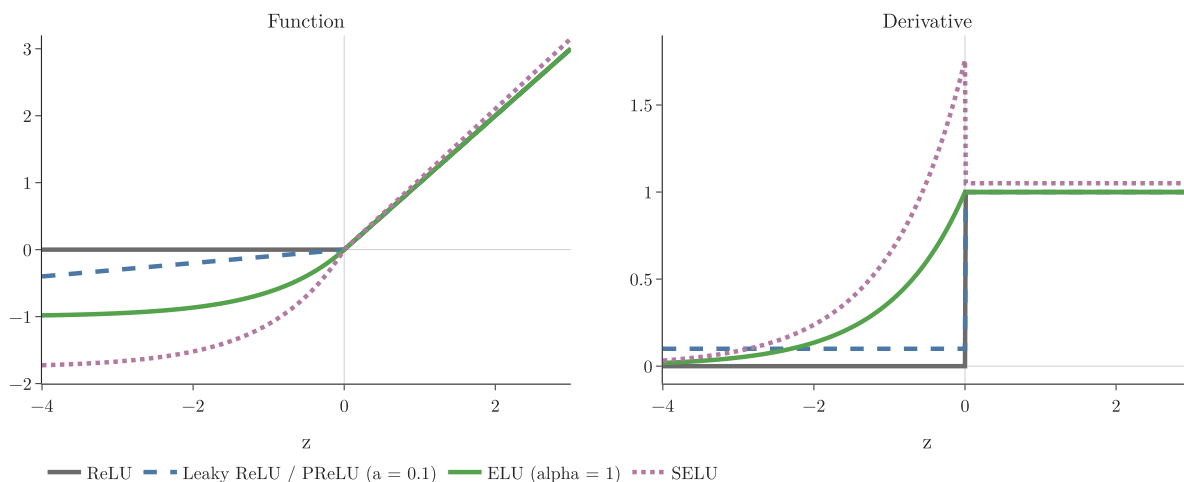


Figure 1: ReLU and its four variants (left) and their derivatives (right). On the negative side ReLU's slope is 0; every variant keeps a slope above 0. The Leaky ReLU slope is drawn as 0.1 instead of 0.01 to make it visible

Figure 1 shows all five functions. The five agree for positive z (SELU is scaled up slightly) and differ only for negative z .

2. Prerequisites

- The [activation functions Note](#): ReLU, saturation, zero-centred outputs.
- The [backpropagation how Note](#): gradients as chain-rule products.

3. The dying ReLU problem

Key point: A dead node outputs 0 for every input. With more than half the nodes dead, the network cannot capture the patterns in the data; with all of them dead, there is no network left.

A **dead neuron** (G-551) is a node whose output is 0 for every input. The dead node's output no longer depends on the input, so it carries no information and learns nothing. Worse, it stays dead for the rest of training: in effect it has been removed from the network.

How much this matters depends on how many nodes die:

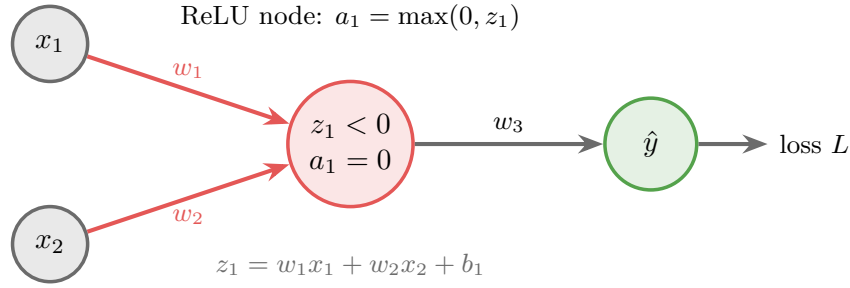
- **A few dead nodes:** the network has less **capacity** (G-344; the range of functions it can fit), but it still works.
- **More than half:** the network runs at less than half its size and cannot represent the patterns in the data well.
- **All of them:** the network outputs a constant. There is no point in training it.

3.1 Why a dead node stops learning

Key point: If $z_1 < 0$, then $a_1 = 0$ and $\partial a_1 / \partial z_1 = 0$. The zero slope appears in the gradient of every weight into the node, so none of them is updated.

Take a small network for regression: two inputs, one hidden ReLU node, one output node (Figure 2). The hidden node computes

$$z_1 = w_1 x_1 + w_2 x_2 + b_1, \quad a_1 = \max(0, z_1)$$



$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

= 0 when $z_1 < 0$, so $\partial L / \partial w_1 = \partial L / \partial w_2 = 0$:
 w_1 and w_2 never change, and z_1 stays negative

Figure 2: A ReLU node with $z_1 < 0$. The factor $\partial a_1 / \partial z_1$ is 0, so the gradients of w_1 and w_2 are 0

Backpropagation (G-247) reaches w_1 through the chain $L \rightarrow \hat{y} \rightarrow a_1 \rightarrow z_1 \rightarrow w_1$:

1. **In words:** the gradient of w_1 is a product of four factors, one of which is ReLU's slope.
2. **Formula:**

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_1}$$

The gradient of w_2 has the same first three factors.

3. **Example:** with $w_1 = -0.8$, $w_2 = -0.5$, $b_1 = 0.1$ and inputs $x_1 = 0.6$, $x_2 = 0.4$:

$$z_1 = -0.48 - 0.20 + 0.1 = -0.58 < 0, \quad \frac{\partial a_1}{\partial z_1} = 0$$

The last factor is $\partial z_1 / \partial w_1 = x_1 = 0.6$. The product for w_1 , factor by factor:

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial a_1} \cdot 0 \cdot 0.6 = 0$$

So $\partial L / \partial w_1 = \partial L / \partial w_2 = 0$, and $w_{\text{new}} = w_{\text{old}} - \eta \cdot 0 = w_{\text{old}}$.

Neither weight changes. If z_1 is negative for every **observation** (G-1374; every record of the data), the node never gets an update again: it is dead.

3.2 What makes z negative

Key point: Two causes: a learning rate so high that one update throws the weights negative, and a large negative bias.

1. A high learning rate (G-1068). Suppose that for the first observation z_1 is positive, so the gradient is not 0. With a very large learning rate, $\eta \partial L / \partial w$ is a big number. Subtracting it from a small weight makes w_1 and w_2 strongly negative, and in the next round z_1 is negative for every observation.

2. A large negative bias. If b_1 is very negative, $z_1 = w_1 x_1 + w_2 x_2 + b_1$ is negative however the inputs vary. The bias can start out negative, or it can be pushed there by updates, again usually because the learning rate is high.

3.3 Why a dead node stays dead

Key point: The weights and bias no longer change, and the scaled inputs are too small to bring z back above 0.

Once z_1 is negative for every observation, nothing can make z_1 positive again:

- the weights and the bias get no updates, so they stay as they are;
- only the **features** (G-772) x_1 , x_2 (the input variables) change from observation to observation, and they are scaled into a small range, so they cannot outweigh a large negative bias or negative weights.

For these two reasons a dead node is called permanently dead.

3.4 Dead nodes in practice

Key point: On the moons data, a learning rate of 10 kills 41% and 69% of the nodes in two hidden layers, and a bias of -1 kills 72% and 100%; accuracy drops to guessing. Leaky ReLU and ELU with the same bias reach 89% and 96%.

The Notebook trains a network with two hidden layers of 32 nodes on 500 standardised observations of `make_moons` (G-111; two features, two classes), with plain **stochastic gradient descent (SGD)** (G-1892) for 200 **epochs** (G-696). The Notebook then counts the nodes whose z is negative for every training observation.

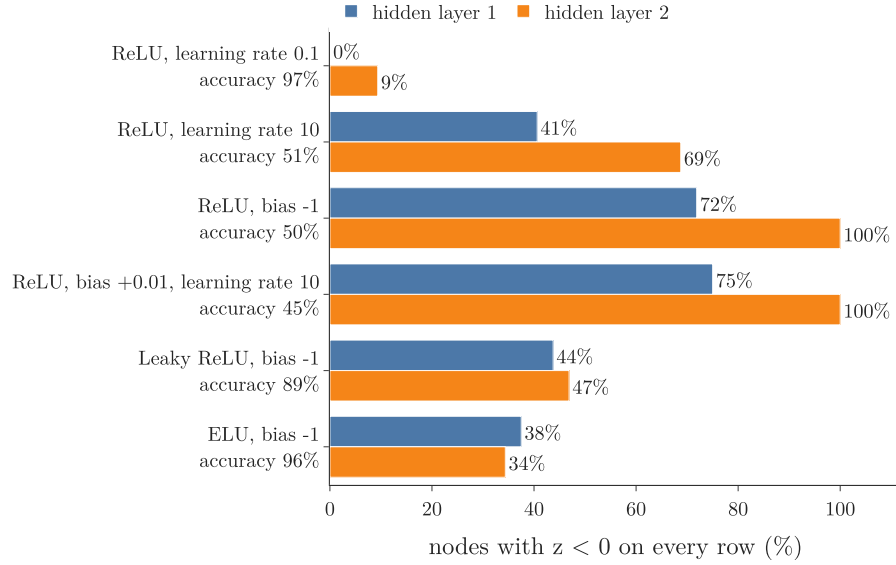


Figure 3: Share of nodes whose z is negative on every training observation, after 200 epochs, with the training accuracy. For ReLU these nodes are dead; for Leaky ReLU and ELU they still pass a gradient

Figure 3 shows the results:

- **ReLU, learning rate 0.1:** almost no dead nodes (0% and 9%), accuracy 97%.
- **ReLU, learning rate 10:** 41% and 69% dead, accuracy 51%. The nodes were already dead after the first epoch; the shares did not change in the 199 epochs after it.
- **ReLU, bias -1 :** 72% and 100% dead from the very start, and still the same after 200 epochs. Accuracy 50%.

The deaths are permanent. Training the learning-rate-10 network for 20 more epochs changes the weights into its 13 dead first-layer nodes by exactly 0, as section 3.1 predicts.

Figure 4 follows the same runs through training, on a log scale so that the first epoch is visible.

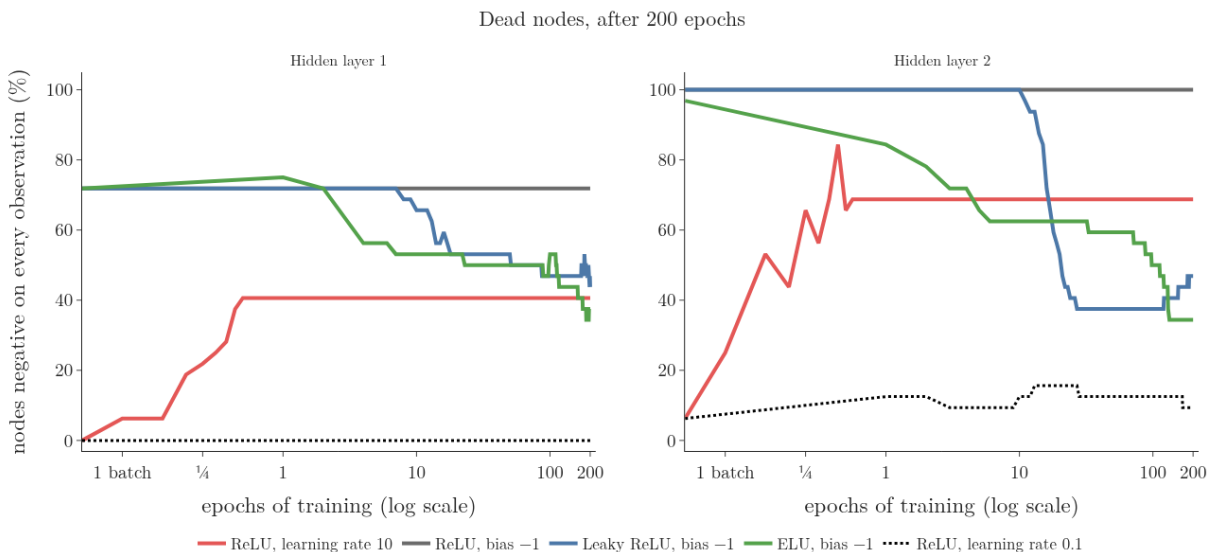


Figure 4: The share of nodes negative on every training observation during training, for the runs of Figure 3. The time axis is logarithmic; the learning-rate-10 run is also shown after each batch of its first epoch.

1. **Learning rate 10 (red).** The nodes die during the very first epoch. After 8 of its 16 batches, 41% of the first-layer nodes are negative everywhere; after that the share never moves again.
2. **Bias -1 with ReLU (grey).** 72% and 100% of the nodes start negative everywhere and stay so for all 200 epochs: a flat line.
3. **Bias -1 with Leaky ReLU (blue) and ELU (green).** The same nodes start negative, but their slope is not 0, so their weights keep changing. The ELU shares start falling within 3 epochs; the Leaky ReLU shares from epoch 8 in the first layer and epoch 11 in the second, where they drop from 100% to 38% by epoch 27.
4. **Learning rate 0.1 (dotted).** Few nodes are ever negative everywhere.

4. Three ways to prevent dead nodes

Key point: Use a lower learning rate, start the biases at a small positive value such as 0.01, or replace ReLU with one of its variants.

1. **A lower learning rate.** In Figure 3, changing only the learning rate, from 10 to 0.1, lowers the dead shares from 41% and 69% to 0% and 9%.
2. **A positive starting bias.** Starting every bias at a small positive value, such as 0.01 or 0.1, makes it very likely that every ReLU node starts active for most inputs (Goodfellow et al. 2016, §6.3.1, suggests 0.1).
3. **A ReLU variant.** The root cause is that ReLU's slope is exactly 0 for $z < 0$. The variants keep everything good about ReLU but give the negative side a slope.

Extra: A positive bias guards the start, not against a learning rate that is too high. Averaged over 5 seeds in the Notebook, a bias of +0.01 with learning rate 10 still ends with 49% and 80% dead nodes (bias 0: 49% and 89%), and accuracy stays near 50%. At learning rate 0.1, a starting bias of +0.1 lowers the share of second-layer nodes that start dead from 5.6% to 3.1%, and the share dead after training from 7.5% to 5.6%.

Python: Setting the starting bias in Keras.

```
keras.layers.Dense(  
    32, activation="relu",  
    bias_initializer=keras.initializers.Constant(0.01))
```

5. Linear variants

Key point: Leaky ReLU and Parametric ReLU replace the 0 on the negative side by a straight line az with a small slope a .

The **linear variants** of ReLU change only the negative side, and change it to a straight line.

5.1 Leaky ReLU

Key point: $f(z) = z$ for $z \geq 0$ and $0.01z$ for $z < 0$. Its slope on the negative side is 0.01 instead of 0, so a small gradient always flows.

1. **In words:** the **Leaky ReLU** (G-1064) keeps positive values and multiplies negative ones by 0.01 (Maas et al. 2013).
2. **Formula:**

$$f(z) = \begin{cases} z & z \geq 0 \\ 0.01z & z < 0 \end{cases} \quad f'(z) = \begin{cases} 1 & z \geq 0 \\ 0.01 & z < 0 \end{cases}$$

3. **Example:** $f(5) = 5$ and $f(-5) = -0.05$. At $z = -5$ ReLU's slope would be 0; Leaky ReLU's is 0.01.

Because $\partial a / \partial z$ is never 0, the gradient in section 3.1 is never exactly 0. The weights keep changing a little, and a node can climb back out of the negative region: with a starting bias of -1 , the share of first-layer nodes negative on every observation falls from 72% to 44% during training. In Figure 3, Leaky ReLU (Keras' "leaky_relu", slope 0.2) with a starting bias of -1 reaches 89% accuracy, where ReLU stayed at 50%.

Advantages:

1. **Non-saturating** (G-1339) on both sides: the output is unbounded in both directions.
2. **Easy to compute:** no exponentials.
3. **No dying ReLU problem.**
4. **Close to zero-centred** (G-2148): outputs can be negative as well as positive, though not symmetric.

The only questionable point of Leaky ReLU is the constant: why 0.01 and not some other value? The value 0.01 was chosen by experiment. Parametric ReLU lets the data choose it instead.

Extra: Keras does not use 0.01 by default (Keras documentation). The string `activation="leaky_relu"` uses slope 0.2, and the layer `keras.layers.LeakyReLU()` uses 0.3. To get 0.01, write `keras.layers.LeakyReLU(negative_slope=0.01)`.

5.2 Parametric ReLU (PReLU)

Key point: Like Leaky ReLU, but the negative slope a is a trainable parameter, learned per node along with the weights.

1. **In words:** the **Parametric ReLU (PReLU)** (G-1452) multiplies negative inputs by a slope a that is learned during training.
2. **Formula:**

$$f(z) = \begin{cases} z & z \geq 0 \\ a z & z < 0 \end{cases}$$

3. **Example:** if training sets $a = 0.25$, then $f(-2) = -0.5$; with $a = 0.01$ it would be Leaky ReLU, with $a = 0$ plain ReLU.

The slope a is a parameter like a weight, found by gradient descent from the data. The slope a is not a **hyperparameter** (G-910) that we set. Everything else, advantages included, is as for Leaky ReLU. The extra flexibility can help: the paper that introduced PReLU reports better **ImageNet** (G-920) accuracy than with ReLU (He et al. 2015).

Python: PReLU is a separate layer after a **Dense** layer without activation.

```
keras.layers.Dense(8),
```

```
keras.layers.PReLU(), # one learnable a per node
```

Here the PReLU layer adds 8 trainable parameters, one slope per node. In Keras each starts at 0, so the network begins as a ReLU network (He et al. 2015 started them at 0.25).

Extra: Nothing keeps the learned slope small. On the moons data the 8 slopes came out between -1.5 and 3.4 . A negative a makes the function V-shaped, which is still a valid non-linear activation.

6. Non-linear variants

Key point: ELU and SELU use an exponential curve for negative z that flattens out at a negative value. Outputs are close to zero-centred and there are no dead nodes.

The **non-linear variants** of ReLU use a curve, not a straight line, on the negative side.

6.1 ELU

Key point: $f(z) = z$ for $z \geq 0$ and $\alpha(e^z - 1)$ for $z < 0$. Smooth, close to zero-centred, no dead nodes, often better test results than ReLU; slower because of the exponential.

1. **In words:** the **ELU** (G-674; exponential linear unit; Clevert et al. 2016) is ReLU for positive z and an exponential curve for negative z that levels off at $-\alpha$.
2. **Formula:**

$$f(z) = \begin{cases} z & z \geq 0 \\ \alpha(e^z - 1) & z < 0 \end{cases} \quad f'(z) = \begin{cases} 1 & z \geq 0 \\ f(z) + \alpha & z < 0 \end{cases}$$

3. **Example:** with $\alpha = 1$ and $z = -1$:

$$f(-1) = e^{-1} - 1 = 0.368 - 1 = -0.632, \quad f'(-1) = -0.632 + 1 = 0.368$$

The negative-side slope comes from differentiating: $\frac{d}{dz} \alpha(e^z - 1) = \alpha e^z = f(z) + \alpha$. A larger α pulls the negative side further down.

Advantages:

1. **Close to zero-centred**, so it converges faster.
2. **Better generalisation** (G-838): in experiments it often gives better results on test data than ReLU (Clevert et al. 2016).
3. **No dying ReLU problem:** the slope is above 0 for every z ($\alpha e^z > 0$ on the negative side). In Figure 3, ELU with a starting bias of -1 reaches 96%.
4. **Continuous and differentiable everywhere** (with $\alpha = 1$, which Figure 1 uses): there is no corner at 0.

Disadvantage: it needs an exponential, so it is slower to compute than ReLU. Faster convergence partly makes up for it, because fewer epochs are needed.

6.2 SELU

Key point: SELU is ELU multiplied by a fixed $\lambda \approx 1.0507$, with fixed $\alpha \approx 1.6733$. Its outputs keep mean 0 and standard deviation 1 from layer to layer: it is self-normalising.

1. **In words:** the **SELU** (G-1766; scaled exponential linear unit; Klambauer et al. 2017) is ELU with a specific α , multiplied by a scale λ .
2. **Formula:**

$$f(z) = \lambda \begin{cases} z & z \geq 0 \\ \alpha(e^z - 1) & z < 0 \end{cases} \quad \lambda \approx 1.0507, \alpha \approx 1.6733$$

3. **Example:**

$$f(1) = 1.0507, \quad f(-1) = 1.0507 \times 1.6733 \times (0.368 - 1) = -1.111$$

λ and α are fixed constants, not trainable parameters. The two constants were derived so that the function has one special property.

The special property is being **self-normalising** (G-1765): the outputs of a SELU layer have mean about 0 and standard deviation about 1, and the next layer keeps them there. Normalised values between layers make the network converge fast, and SELU also generalises well in experiments.

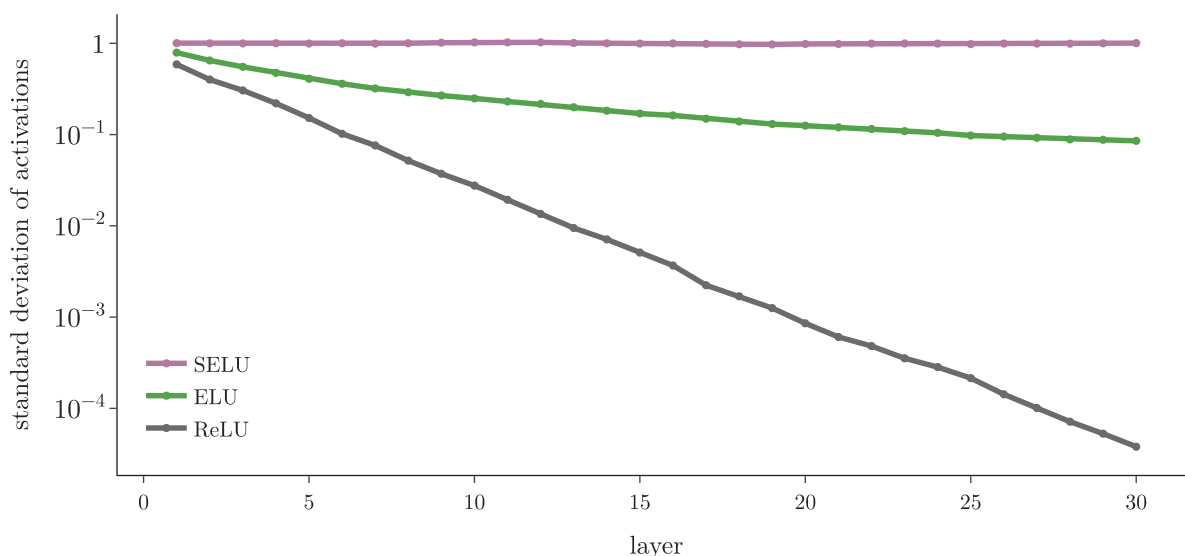


Figure 5: Standard deviation of the activations through 30 layers of 256 nodes, with no training. SELU stays at 1; ELU and ReLU shrink layer after layer

Figure 5 shows this in the Notebook. Standard-normal inputs pass through 30 layers with the same random weights for each activation. With SELU the standard deviation is 1.00 at layer 1 and still 1.00 at layer 30. With ReLU it is 0.59 at layer 1 and 4×10^{-5} at layer 30.

The disadvantage of SELU is adoption. SELU is used in few places so far, for three reasons:

- SELU is recent (2017);
- its paper has 9 pages plus a 93-page appendix of proofs;
- less research builds on it.

Extra: Self-normalisation rests on assumptions in the paper: inputs with mean 0 and variance 1, and weights drawn with variance 1/inputs (Klambauer et al. 2017). Keras' documentation for `selu` therefore asks for `kernel_initializer="lecun_normal"` (as in Figure 5) and for `AlphaDropout` instead of ordinary dropout, which the paper shows disturbs the mean and variance. Starting weights are the subject of the [weight initialisation Note](#).

Python: ELU and SELU in Keras.

```
keras.layers.Dense(32, activation="elu")
```

```
keras.layers.Dense(32, activation="selu",
                    kernel_initializer="lecun_normal")
```

6.3 GELU and SiLU: smooth versions of ReLU

Key point: GELU is $z \Phi(z)$ and SiLU is $z \sigma(z)$: the input times a number between 0 and 1 that grows with the input. Both look like ReLU with a smooth bend and a small dip below 0. Transformers use them.

ReLU makes a hard choice: it keeps z when z is positive and replaces it by 0 when z is negative. A softer rule makes the same choice at random, with odds that depend on z :

1. **Keep or drop.** The node's value z is kept with probability $p(z)$ and replaced by 0 otherwise. This is [dropout](#) (G-639) with a rate that depends on the input.
2. **The odds follow the input.** A large positive z is almost always kept; a very negative z is almost always dropped.
3. **Take the average.** Over many random choices the average output is $z \times p(z) + 0 \times (1 - p(z)) = z p(z)$. Using this average as the activation function needs no randomness at all.

The **GELU** (G-834; Gaussian error linear unit) uses for p the standard normal **cumulative distribution function** (G-515) $\Phi(z)$ (G-18): the probability that a standard normal value is below z . For example, $\Phi(0) = 0.50$ because half of the values are below 0, and $\Phi(1) = 0.84$ because 84 percent are below 1.

1. **In words:** the input times the probability of keeping it.
2. **Formula:**

$$\text{GELU}(z) = z \Phi(z)$$

3. **Example:**

z	-2	-1	0	1	2
$\Phi(z)$, probability of being kept	0.02	0.16	0.50	0.84	0.98
GELU, $z \Phi(z)$	-0.05	-0.16	0	0.84	1.95

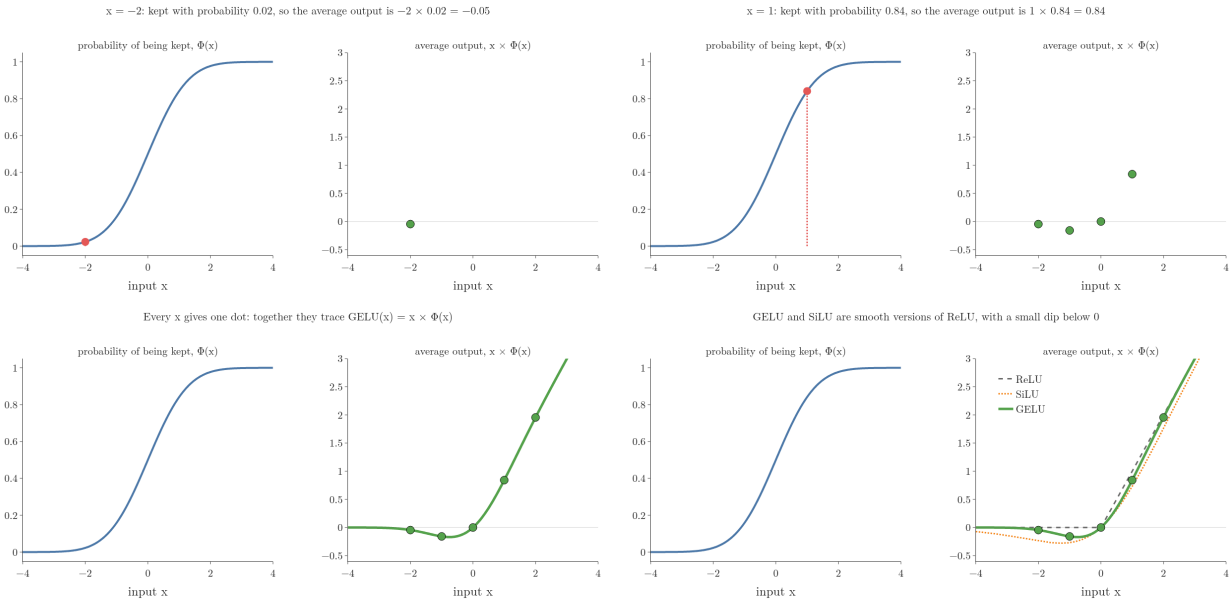


Figure 6: GELU built point by point. Left: the probability $\Phi(z)$ of keeping the input. Right: the average output $z \Phi(z)$, one dot per input, then the whole curve, then ReLU and SiLU for comparison. Idea after StatQuest, “The GELU, SiLU and SwiGLU activation functions, clearly explained!!!”

Figure 6 builds the curve from these five values. Watch the red dot on the left: its height is the number that multiplies the input on the right.

The **SiLU** (G-2266) (sigmoid linear unit, also called **Swish**) uses the **sigmoid** (G-1798) for p , a curve of almost the same shape as Φ : $\text{SiLU}(z) = z \sigma(z)$. For example $\text{SiLU}(-1) = -1 \times 0.269 = -0.27$ and $\text{SiLU}(1) = 0.73$.

Compared with ReLU (last frame of Figure 6):

- **For large positive z** both are almost z , like ReLU, so they do not saturate there.
- **For negative z** the output is a small negative number, not 0: GELU dips to -0.17 and returns towards 0. The slope is not exactly 0, so a node with a slightly negative z still receives a gradient.
- **At 0** the curve bends smoothly; there is no corner.

GELU is the activation inside GPT’s feed-forward layers; its use there is in section 6.4 of the [GPT Note](#).

7. Summary

	ReLU	Leaky ReLU	PReLU	ELU	SELU
Negative side Kind	0	$0.01z$ linear variant	az , a learned linear variant	$\alpha(e^z - 1)$ non-linear variant	$\lambda\alpha(e^z - 1)$ non-linear variant
Slope for $z < 0$	0	0.01	a	$f(z) + \alpha$	$\lambda\alpha e^z$
Dead nodes	yes	no	no	no	no
Zero-centred	no	close	close	close	yes (self-normalising)
Exponential Main drawback	no dying	no slope 0.01 is arbitrary	no	yes slower	yes not widely adopted

- A ReLU node with $z < 0$ for every observation has slope 0, gets no updates and is dead for good.
- Causes: a high learning rate and a large negative bias. Fixes: a lower learning rate, a bias starting at 0.01, or a variant.
- Linear variants (Leaky ReLU, PReLU) put a straight line on the negative side; non-linear ones (ELU, SELU) a curve.
- SELU keeps activations at mean 0 and standard deviation 1 across layers.
- GELU, $z\Phi(z)$, and SiLU, $z\sigma(z)$, are smooth versions of ReLU: the input times a keep-probability that grows with the input.

8. Sources

Built from

- CampusX, “Relu Variants Explained | Leaky Relu | Parametric Relu | Elu | Selu | Activation Functions Part 2”, YouTube, <https://www.youtube.com/watch?v=2OwWs7Hzr9g>
- StatQuest with Josh Starmer, “The GELU, SiLU and SwiGLU activation functions, clearly explained!!!”, YouTube, <https://www.youtube.com/watch?v=2FaI2Fen1mQ> (section 6.3: GELU and SiLU as the average of input-dependent dropout, Figure 6)

Other references

- Clevert, D.-A., Unterthiner, T. and Hochreiter, S. (2016). Fast and Accurate Deep Network Learning by Exponential Linear Units (ELUs). ICLR 2016. arXiv:1511.07289.
- He, K., Zhang, X., Ren, S. and Sun, J. (2015). Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification. ICCV 2015. arXiv:1502.01852.
- Goodfellow, I., Bengio, Y. and Courville, A. (2016). *Deep Learning*, MIT Press, section 6.3.1 (a small positive starting bias keeps ReLU units active at the start).
- Keras API documentation: `LeakyReLU`, `PReLU`, `selu` (keras.io/api).
- Klambauer, G., Unterthiner, T., Mayr, A. and Hochreiter, S. (2017). Self-Normalizing Neural Networks. NeurIPS 2017. arXiv:1706.02515.

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9. Key terms

Term	Meaning
Dying ReLU problem	ReLU nodes ending up with a negative weighted sum for every input, so they output 0 and stop learning
Observation	One record of the data: one row of the data table
Feature	An input variable, such as x_1 : one column of the data table
Dead neuron	A node whose output is 0 for every input; it gets no updates and stays that way
Linear variants of ReLU	ReLU variants with a straight line on the negative side: Leaky ReLU and PReLU
Non-linear variants of ReLU	ReLU variants with a curve on the negative side: ELU and SELU
Leaky ReLU	z for $z \geq 0$, $0.01z$ for $z < 0$
Parametric ReLU (PReLU)	Leaky ReLU whose negative slope a is learned during training, one per node
ELU	Exponential linear unit: z for $z \geq 0$, $\alpha(e^z - 1)$ for $z < 0$
SELU	Scaled ELU, $\lambda \approx 1.0507$ times ELU with $\alpha \approx 1.6733$; self-normalising
GELU	Gaussian error linear unit, $z \Phi(z)$: the input times the standard normal probability of a value below it; a smooth version of ReLU
SiLU (Swish) (G-2266)	Sigmoid linear unit, $z \sigma(z)$: the input times its sigmoid; close to GELU
Self-normalising	Keeping the activations of every layer at mean 0 and standard deviation 1 without a separate normalisation step